

SKILL 24

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("=== Simple Shell Demo ===\n");

    while (1) {
        printf("> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Exit\n");
            break;
        }

        if (strcmp(input, "pipeline") == 0) {
            printf("Pipeline demo: ls | wc -l\n");
        }
        else if (strcmp(input, "signal") == 0) {
            printf("Signal handling demo executed\n");
        }
        else if (strcmp(input, "jobs") == 0) {
            printf("Job control demo executed\n");
        }
        else {
            printf("Processed input: %s\n", input);
        }
    }

    return 0;
}
```

nano Makefile:

```
CC=gcc
CFLAGS=-Wall -Wextra

all:
    $(CC) $(CFLAGS) src/main.c -o shell_demo

clean:
    rm -f shell_demo

run:
    ./shell_demo
```

nano README.md:

```
# Mini Shell Project_
```

Description

This project demonstrates basic shell concepts developed through the previous tasks.

Features demonstrated:

- Command input
- Exit command
- Pipelines
- Signal handling
- Job control
- Error handling
- Process management

Build

Run:

```
make
```

Execute

Run:

```
./shell_demo
```

or:

```
make run
```

Demo Commands

```
pipeline
```

```
signal
```

```
jobs
```

```
hello
```

```
exit
```

Project Structure

task24/

- |
- | -- src/
- | | -- main.c
- |
- | -- docs/
- |
- | -- README.md
- |
- | -- Makefile

Final Review

The project structure, build process, features, and demonstration scenarios were verified.

```
nishanth@LAPTOP-6Q3ON07H:~$ mkdir task24
nishanth@LAPTOP-6Q3ON07H:~$ cd task24
nishanth@LAPTOP-6Q3ON07H:~/task24$ mkdir src docs
nishanth@LAPTOP-6Q3ON07H:~/task24$ touch README.md
nishanth@LAPTOP-6Q3ON07H:~/task24$ touch Makefile
nishanth@LAPTOP-6Q3ON07H:~/task24$ nano src/main.c
nishanth@LAPTOP-6Q3ON07H:~/task24$ nano Makefile
nishanth@LAPTOP-6Q3ON07H:~/task24$ nano README.md
nishanth@LAPTOP-6Q3ON07H:~/task24$ make
gcc -Wall -Wextra src/main.c -o shell_demo
nishanth@LAPTOP-6Q3ON07H:~/task24$ ./shell_demo
=== Simple Shell Demo ===
> pipeline
signal
jobs
hello
exitPipeline demo: ls | wc -l
> Signal handling demo executed
> Job control demo executed
> Processed input: hello
>
Exit
nishanth@LAPTOP-6Q3ON07H:~/task24$ find . -maxdepth 2 -type f
./README.md
./src/main.c
./shell_demo
./Makefile
nishanth@LAPTOP-6Q3ON07H:~/task24$ make clean
rm -f shell_demo
nishanth@LAPTOP-6Q3ON07H:~/task24$ make
gcc -Wall -Wextra src/main.c -o shell_demo
```

```
nishanth@LAPTOP-6Q30N07H:~/task24$ make
gcc -Wall -Wextra src/main.c -o shell_demo
nishanth@LAPTOP-6Q30N07H:~/task24$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/nishanth/task24/.git/
nishanth@LAPTOP-6Q30N07H:~/task24$ git add .
nishanth@LAPTOP-6Q30N07H:~/task24$ git commit -m "Final shell project"
[master (root-commit) 5b93a2e] Final shell project
 4 files changed, 104 insertions(+)
 create mode 100644 Makefile
 create mode 100644 README.md
 create mode 100755 shell_demo
 create mode 100644 src/main.c
nishanth@LAPTOP-6Q30N07H:~/task24$ git tag v1.0
nishanth@LAPTOP-6Q30N07H:~/task24$ git tag
v1.0
nishanth@LAPTOP-6Q30N07H:~/task24$ git status
On branch master
nothing to commit, working tree clean
nishanth@LAPTOP-6Q30N07H:~/task24$
```